

Digital Transformation — Module 04: Cognition —

Discussion Questions

20 questions on algorithmic decision-making, AI governance, explainability, and ethics.

1. How should AI governance differ for high-impact decisions (hiring, lending) versus low-impact ones (product recommendations)?
2. The COMPAS case revealed racial bias in algorithmic risk scoring. How should organizations test for and prevent such bias?
3. When is an opaque but accurate algorithm preferable to an explainable but less accurate one?
4. How do algorithmic decisions redistribute power within organizations?
5. When should human judgment override algorithmic recommendations?
6. How should organizations handle the "accountability gap" — when nobody can fully explain why an algorithm made a decision?
7. Compare rule-based systems (explicit, explainable) with ML-based systems (learned, often opaque) for organizational decision-making.
8. How do you design a "kill switch" for algorithmic decisions that discovers the system is producing harmful outputs?
9. What training do managers need to effectively oversee algorithmic decision systems?
10. When should an organization refuse to automate a decision even if it could improve accuracy?
11. Design an AI governance framework for your organization. Include accountability, transparency, fairness, and oversight components.
12. Identify three decisions in your organization that are or could be made algorithmically. For each: what governance is needed?
13. Design an "algorithmic audit" process that periodically reviews automated decisions for bias and accuracy.
14. How should organizations communicate to stakeholders when decisions affecting them are made by algorithms?

15. Compare how regulated industries (finance, healthcare) and unregulated industries approach algorithmic accountability differently.
16. Design an explainability interface that helps non-technical decision-makers understand algorithmic recommendations.
17. How does the EU AI Act and similar regulation change organizational AI governance requirements?
18. When multiple algorithms interact (recommendation feeds into pricing feeds into inventory), how does governance scale?
19. Design a process for handling complaints about algorithmic decisions to ensure fairness and recourse.
20. How will increasing AI capability change the human-algorithm governance relationship over time?